

HW3 - STAT 709

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Problem 1. Textbook exercises

Jun Shao's book, Chapter 1: Exercises 60 and 82 (required); Exercise 91 (optional).

Ex. 1.60

Let X be a random variable with $\mathbb{E}X^2 < \infty$ and let $Y = |X|$. Suppose that X has a Lebesgue p.d.f. symmetric about 0. Show that X and Y are uncorrelated, but they are not independent.

Sol.

Notice that

$$I := \int_0^\infty xf(x)dx = - \int_{-\infty}^0 xf(x)dx$$

and hence $\mathbb{E}X = 0$ and $\mathbb{E}Y = 2I$. Assume f to be the p.d.f. of X , then

$$\begin{aligned}\text{Cov}(X, Y) &= \mathbb{E}XY \\ &= \int_0^\infty x^2 f(x)dx - \int_{-\infty}^0 x^2 f(x)dx = 0\end{aligned}$$

since $f(x) = f(-x)$ for any $x \in \mathbb{R}$, which means X and Y are uncorrelated.

Notice that

$$P(X \geq a, Y \geq a) = P(X \geq a), \quad P(Y \geq a) = 2P(X \geq a)$$

and if X and Y are independent, then we have

$$2P(X \geq a)^2 = P(X \geq a),$$

which means that $P(X \geq a) = 1/2$ or 0 for any $a \in \mathbb{R}$, which is a contradiction by the continuity of the tail probability.

Ex. 1.82

Let (X_1, X_2) be $N_k(\mu, \Sigma)$ with a $k \times k$ positive definite

$$\Sigma = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix},$$

where X_1 is a random l -vector and Σ_{11} is an $l \times l$ matrix. Show that the conditional Lebesgue p.d.f. of X_2 given $X_1 = x_1$ is

$$N_{k-l}(\mu_2 + \Sigma_{21}\Sigma_{11}^{-1}(x_1 - \mu_1), \Sigma_{22} - \Sigma_{21}\Sigma_{11}^{-1}\Sigma_{12}),$$

where $\mu_i = \mathbb{E}X_i$, $i = 1, 2$.

Sol.

We would like to prove that

$$\mu((\mathbf{x}_1, \mathbf{x}_2), A) := \int_A \frac{1}{(2\pi)^{(k-l)/2} (\det(\Sigma'))^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x} - \mu'(\mathbf{x}_1))^T \Sigma'^{-1}(\mathbf{x} - \mu'(\mathbf{x}_1))\right) d\mathbf{x}$$

for $\mathbf{x}_1 \in \mathbb{R}^l$, $\mathbf{x}_2 \in \mathbb{R}^{k-l}$ and where

$$\Sigma' := \Sigma_{22} - \Sigma_{21}\Sigma_{11}^{-1}\Sigma_{12}, \quad \mu'(\mathbf{x}_1) := \mu_2 + \Sigma_{21}\Sigma_{11}^{-1}(\mathbf{x}_1 - \mu_1)$$

is a regular conditional distribution from $\mathbb{R}^k \times \mathcal{B}^{k-l} \rightarrow [0, 1]$. It remains to show that $\mu(\cdot, A)$ is a version of $P(X_2 \in A | X_1)$, and notice

$$\begin{aligned} & \int_{\{X_1 \leq a\}} P(X_2 \in A | X_1) dP \\ &= \int_{\mathbf{x}_1 \leq a} \int_{\mathbf{x}_2 \in A} \frac{1}{(2\pi)^{k/2} (\det \Sigma)^{1/2}} \exp\left(-\frac{1}{2}((\mathbf{x}_1, \mathbf{x}_2) - (\mu_1, \mu_2))^T \Sigma^{-1}((\mathbf{x}_1, \mathbf{x}_2) - (\mu_1, \mu_2))\right) d\mathbf{x}_1 d\mathbf{x}_2 \\ &= \int_{\mathbf{x}_1 \leq a} \int_A \frac{1}{(2\pi)^{(k-l)/2} (\det(\Sigma'))^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x} - \mu'(\mathbf{x}_1))^T \Sigma'^{-1}(\mathbf{x} - \mu'(\mathbf{x}_1))\right) d\mathbf{x} \\ & \quad \frac{1}{(2\pi)^{l/2} (\det \Sigma_{11})^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x}_1 - \mu_1)^T \Sigma_{11}^{-1}(\mathbf{x}_1 - \mu_1)\right) d\mathbf{x}_1 \\ &= \int \int_{\mathbf{x}_1 \leq a} \int_A \frac{1}{(2\pi)^{(k-l)/2} (\det(\Sigma'))^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x} - \mu'(\mathbf{x}_1))^T \Sigma'^{-1}(\mathbf{x} - \mu'(\mathbf{x}_1))\right) d\mathbf{x} \\ & \quad \frac{1}{(2\pi)^{k/2} (\det \Sigma)^{1/2}} \exp\left(-\frac{1}{2}((\mathbf{x}_1, \mathbf{x}_2) - (\mu_1, \mu_2))^T \Sigma^{-1}((\mathbf{x}_1, \mathbf{x}_2) - (\mu_1, \mu_2))\right) d\mathbf{x}_1 d\mathbf{x}_2 \\ &= \int_{\{X_1 \leq a\}} \mu(\cdot, A) dP \end{aligned}$$

where

$$\det \Sigma' \det \Sigma_{11} = \det \Sigma$$

by Schur's lemma and that

$$\Sigma^{-1} = \begin{pmatrix} \Sigma_{11}^{-1} + \Sigma_{11}^{-1}\Sigma_{12}\Sigma'^{-1}\Sigma_{21}\Sigma_{11}^{-1} & -\Sigma_{11}^{-1}\Sigma_{12}\Sigma'^{-1} \\ -\Sigma'^{-1}\Sigma_{21}\Sigma_{11}^{-1} & \Sigma'^{-1} \end{pmatrix},$$

for any $a \in \mathbb{R}^l$, where $\{\mathbf{x} \leq a\}$ is the rectangle defined by the inequality coordinate-wise and Σ' is positive definite, and we are done.

Ex. 1.91 (optional)

Let X , Y , and Z be random variables on a probability space. Suppose that $\mathbb{E}|X| < \infty$ and $Y = h(Z)$ with a Borel h . Show that

- (a) If X and Z are independent and $\mathbb{E}|Z| < \infty$, then $\mathbb{E}(XZ | Y) = \mathbb{E}(X)\mathbb{E}(Z | Y)$ a.s.
- (b) If $\mathbb{E}[f(X) | Z] = f(Y)$ for all bounded continuous functions f on $\mathbb{R}$, then $X = Y$ a.s.
- (c) If $\mathbb{E}[f(X) | Z] \geq f(Y)$ for all bounded, continuous, nondecreasing functions f on $\mathbb{R}$, then $X \geq Y$ a.s.

Sol.

- (a) Independence gives $\mathbb{E}|XZ| = \mathbb{E}|X|\mathbb{E}|Z| < \infty$. For every Borel set $A \subseteq \mathbb{R}$, the random variable $Z\mathbf{1}_{\{Y \in A\}} = Z\mathbf{1}_{\{h(Z) \in A\}}$ is an integrable function of Z , and hence is independent of X . Therefore

$$\begin{aligned}\mathbb{E}[XZ\mathbf{1}_{\{Y \in A\}}] &= \mathbb{E}X \mathbb{E}[Z\mathbf{1}_{\{Y \in A\}}] \\ &= \mathbb{E}[\mathbb{E}X \mathbb{E}[Z | Y]\mathbf{1}_{\{Y \in A\}}].\end{aligned}$$

Since $\mathbb{E}X \mathbb{E}[Z | Y]$ is integrable and $\sigma(Y)$ -measurable, it is a version of $\mathbb{E}[XZ | Y]$.

- (b) Put $u(x) = \arctan x$. Both u and u^2 are bounded and continuous, so the assumption implies

$$\mathbb{E}[u(X) | Z] = u(Y), \quad \mathbb{E}[u(X)^2 | Z] = u(Y)^2.$$

Since $Y = h(Z)$, $u(Y)$ is $\sigma(Z)$ -measurable. Thus

$$\begin{aligned}\mathbb{E}[(u(X) - u(Y))^2 | Z] &= \mathbb{E}[u(X)^2 | Z] - 2u(Y)\mathbb{E}[u(X) | Z] + u(Y)^2 \\ &= 0.\end{aligned}$$

Taking expectations shows $u(X) = u(Y)$ a.s. As u is strictly increasing, $X = Y$ a.s.

- (c) Fix $r \in \mathbb{Q}$ and define $f_n(x) = \min\{1, n(x-r)_+\}$, where $(x-r)_+ = \max\{x-r, 0\}$. These functions are bounded, continuous and nondecreasing, and $f_n(x) \uparrow \mathbf{1}_{\{x > r\}}$. Conditional monotone convergence and the assumption give

$$\mathbb{P}(X > r | Z) \geq \mathbf{1}_{\{Y > r\}} \quad \text{a.s.}$$

Since the left side is at most one and $\{Y > r\} \in \sigma(Z)$,

$$\mathbb{P}(X \leq r < Y) = \mathbb{E}[\mathbf{1}_{\{Y > r\}}(1 - \mathbb{P}(X > r | Z))] = 0.$$

Finally, $\{X < Y\} = \bigcup_{r \in \mathbb{Q}} \{X \leq r < Y\}$, so $\mathbb{P}(X < Y) = 0$, proving $X \geq Y$ a.s.

Problem 2. Textbook exercises

Durrett's *Probability: Theory and Examples*, fifth edition: Exercises 2.1.4 and 2.1.6.

Ex. 2.1.4

Let $\Omega = (0, 1)$, $\mathcal{F} =$ Borel sets, $P =$ Lebesgue measure. $X_n(\omega) = \sin(2\pi n\omega)$, $n = 1, 2, \dots$ are uncorrelated but not independent.

Sol.

Notice that

$$\begin{aligned}\text{Cov}(X_m, X_n) &= \int_0^1 \sin(2\pi m\omega) \sin(2\pi n\omega) d\omega \\ &= \frac{1}{2} \left(\int_0^1 \cos(2\pi(m-n)\omega) d\omega - \int_0^1 \cos(2\pi(m+n)\omega) d\omega \right) \\ &= 0\end{aligned}$$

for $m \neq n$, and hence they are uncorrelated. To see they are not independent, we may notice that

$$P(X_n \leq -1/2, X_{2n} \leq -1/2) = 1/8 \neq 1/9 = P(X_n \leq -1/2)P(X_{2n} \leq -1/2)$$

for $n \geq 1$ and we are done.

Ex. 2.1.6

Prove directly from the definition that if X and Y are independent and f and g are measurable functions then $f(X)$ and $g(Y)$ are independent.

Sol.

For any $A, B \in \mathcal{B}(\mathbb{R})$, we know

$$\begin{aligned} P(f(X) \in A, g(Y) \in B) &= P(X \in f^{-1}(A), Y \in g^{-1}(B)) \\ &= P(X \in f^{-1}(A))P(Y \in g^{-1}(B)) \\ &= P(f(X) \in A)P(g(Y) \in B) \end{aligned}$$

and we are done.

The remaining exercises concern conditional expectation and are optional. Read the questions and try as many parts as you can.

Problem 3. Conditioning on a transformation (optional)

Let $X \in \mathbb{R}^n$ have a continuous density p , and let $f, g : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be continuous, with $\mathbb{E} \|f(X)\|_2 < \infty$. Let $A_1, \dots, A_m$ be pairwise disjoint open sets whose union has a Lebesgue-null complement. On each A_j , assume that g is continuously differentiable and one-to-one, with $\det Dg(x) \neq 0$. Write $V_j = g(A_j)$ and let $h_j : V_j \rightarrow A_j$ be its inverse.

We want a formula for $\mathbb{E}[f(X) \mid g(X) = y]$. In general, $(f(X), g(X))$ does not have a joint density on $\mathbb{R}^{2n}$, so the elementary joint-density ratio formula need not apply directly.

For the following local calculation, fix $y \notin \bigcup_{j=1}^m \partial V_j$. Put $J(y) = \{j : y \in V_j\}$, and assume that $p(h_j(y)) > 0$ for at least one $j \in J(y)$. These conditions ensure that a sufficiently small cube around y meets exactly the active inverse branches and that the limiting denominator is positive. For $\Delta > 0$, let $E_\Delta = \prod_{i=1}^n (y_i - \Delta, y_i + \Delta)$.

- (1) For $j \in J(y)$ and sufficiently small Δ , define $B_\Delta^{(j)} = h_j(E_\Delta)$. Prove that

$$\text{Vol}(B_\Delta^{(j)}) = |\det Dh_j(y)| (2\Delta)^n + o(\Delta^n) \quad (\Delta \downarrow 0).$$

- (2) Use part (1) to evaluate

$$\lim_{\Delta \downarrow 0} \frac{\mathbb{E}[f(X) \mathbf{1}\{g(X) \in E_\Delta\}]}{\mathbb{P}(g(X) \in E_\Delta)}.$$

Derive a version of $\mathbb{E}[f(X) \mid g(X) = y]$ in terms of the inverse branches. Explain why this formula, summed over the branches defined at y , gives a conditional expectation for almost every y under the law of $g(X)$.

Sol.

- (1) By the inverse function theorem, V_j is open and h_j is C^1 on V_j . In particular, $|\det Dh_j|$ is continuous.

We know

$$\begin{aligned} \text{Vol}(B_\Delta^{(j)}) &= \int_{h_j(E_\Delta)} d\mathbf{x} \\ &= \int_{E_\Delta} |\det Dh_j(\mathbf{y})| d\mathbf{y} \\ &= |\det Dh_j(y)| (2\Delta)^n + \int_{E_\Delta} (|\det Dh_j(\mathbf{y})| - |\det Dh_j(y)|) d\mathbf{y} \end{aligned}$$

where

$$\begin{aligned} &\lim_{\Delta \rightarrow 0} \left| \int_{E_\Delta} (|\det Dh_j(\mathbf{y})| - |\det Dh_j(y)|) d\mathbf{y} \right| / (\Delta)^n \\ &\leq 2^n \lim_{\Delta \rightarrow 0} \sup_{E_\Delta} ||\det Dh_j(\mathbf{y})| - |\det Dh_j(y)|| = 0 \end{aligned}$$

and we are done by the continuity of $|\det Dh_j|$ at y .

- (2) In the following local calculation, write $J = J(y)$. Since X has a density, the complement of $\bigcup_j A_j$

has probability zero. For sufficiently small Δ , only the branches in $J(y)$ contribute. Notice that

$$\begin{aligned}\mathbb{E}[f(X)\mathbf{1}\{g(X) \in E_\Delta\}] &= \int_{\bigcup_{j \in J} h_j(E_\Delta)} f(\mathbf{x})p(\mathbf{x})d\mathbf{x} \\ &= \sum_{j \in J} \int_{E_\Delta} f(h_j(\mathbf{y}))p(h_j(\mathbf{y}))|\det Dh_j|(\mathbf{y})d\mathbf{y} \\ &= \sum_{j \in J} f(h_j(y))p(h_j(y))|\det Dh_j(y)|(2\Delta)^n + o(\Delta^n)\end{aligned}$$

with a similar proof in (1) and

$$\mathbb{P}(g(X) \in E_\Delta) = \int_{\bigcup_{j \in J} h_j(E_\Delta)} p(\mathbf{x})d\mathbf{x} = \sum_{j \in J} p(h_j(y))|\det Dh_j(y)|(2\Delta)^n + o(\Delta^n)$$

and hence

$$\begin{aligned}\lim_{\Delta \downarrow 0} \frac{\mathbb{E}[f(X)\mathbf{1}\{g(X) \in E_\Delta\}]}{\mathbb{P}(g(X) \in E_\Delta)} &= \lim_{\Delta \downarrow 0} \frac{\sum_{j \in J} f(h_j(y))p(h_j(y))|\det Dh_j(y)|(2\Delta)^n + o(\Delta^n)}{\sum_{j \in J} p(h_j(y))|\det Dh_j(y)|(2\Delta)^n + o(\Delta^n)} \\ &= \frac{\sum_{j \in J} f(h_j(y))p(h_j(y))|\det Dh_j(y)|}{\sum_{j \in J} p(h_j(y))|\det Dh_j(y)|}\end{aligned}$$

which gives the claimed local limit. The denominator is positive by the hypothesis on $p(h_j(y))$ and the invertibility of $Dh_j(y)$.

To define a version globally, put

$$\begin{aligned}q(y) &= \sum_{j \in J(y)} p(h_j(y))|\det Dh_j(y)|, \\ r(y) &= \sum_{j \in J(y)} f(h_j(y))p(h_j(y))|\det Dh_j(y)|.\end{aligned}$$

Each summand is defined on V_j and extended by zero outside V_j , so q and r are Borel measurable. Change of variables on each A_j gives, for every Borel set $A \subseteq \mathbb{R}^n$,

$$\mathbb{P}(g(X) \in A) = \int_A q(y) dy, \quad \mathbb{E}[f(X)\mathbf{1}\{g(X) \in A\}] = \int_A r(y) dy.$$

In particular, q is the density of $g(X)$. Define $H(y) = r(y)/q(y)$ when $q(y) > 0$, and $H(y) = 0$ otherwise. When $q(y) = 0$, all nonnegative branch weights vanish, so $r(y) = 0$ as well. Moreover,

$$\begin{aligned}\mathbb{E} \|H(g(X))\|_2 &= \int_{\{q>0\}} \|r(y)\|_2 dy \\ &\leq \sum_{j=1}^m \int_{V_j} \|f(h_j(y))\|_2 p(h_j(y))|\det Dh_j(y)| dy \\ &= \mathbb{E} \|f(X)\|_2 < \infty.\end{aligned}$$

Thus $H(g(X))$ is integrable and $\sigma(g(X))$ -measurable, and the defining identity follows from

$$\mathbb{E}[H(g(X))\mathbf{1}\{g(X) \in A\}] = \int_A H(y)q(y) dy = \int_A r(y) dy.$$

This global argument does not require excluding branch boundaries: it sums all branches defined at each y , and $\mathbb{P}(q(g(X)) = 0) = 0$. All vector-valued identities are interpreted componentwise.

Problem 4. Conditional independence (optional)

Let X, Y, Z be real-valued random variables. Assume that X and Y are conditionally independent given Z , in the sense that

$$\mathbb{P}(B \mid X, Z) = \mathbb{P}(B \mid Z) \quad \text{a.s. for every } B \in \sigma(Y).$$

The first two parts connect this definition with the familiar density formulation.

- (1) Let (U, W) be random vectors with a joint density. For a Borel function φ satisfying $\mathbb{E}|\varphi(U)| < \infty$, show that a version is

$$\mathbb{E}[\varphi(U) \mid W = w] = \int \varphi(u)p(u \mid w) du$$

for almost every w under the law of W , where $p(u \mid w) = p_{U,W}(u, w)/p_W(w)$ when $p_W(w) > 0$.

- (2) If (X, Y, Z) has a joint density, show that

$$f_{X,Y|Z}(x, y \mid z) = f_{X|Z}(x \mid z)f_{Y|Z}(y \mid z)$$

for almost every z under the law of Z and Lebesgue-almost every (x, y) .

- (3) Without assuming a joint density, prove that, for every $A \in \sigma(X)$ and $B \in \sigma(Y)$,

$$\mathbb{P}(A \cap B \mid Z) = \mathbb{P}(A \mid Z)\mathbb{P}(B \mid Z) \quad \text{a.s.}$$

Hint: Use the tower property and the definition of conditional expectation.

- (4) Let $\mathbf{X} = (X_1, \dots, X_p)^\top \sim N_p(0, \Sigma)$, with Σ positive definite, and define the precision matrix $\Theta = \Sigma^{-1}$. Set $Y = X_1$ and $Z = (X_2, \dots, X_p)^\top$. In the coordinate order (Y, Z) , write

$$\Sigma = \begin{pmatrix} \sigma_{YY} & \sigma_{ZY}^\top \\ \sigma_{ZY} & \Sigma_{ZZ} \end{pmatrix}.$$

Recall that

$$Y \mid Z = z \sim N\left(z^\top \Sigma_{ZZ}^{-1} \sigma_{ZY}, \sigma_{YY} - \sigma_{ZY}^\top \Sigma_{ZZ}^{-1} \sigma_{ZY}\right), \quad \theta_{ZY} = -\theta_{Y Y} \Sigma_{ZZ}^{-1} \sigma_{ZY}.$$

In a Gaussian graphical model, distinct nodes are joined when the corresponding precision-matrix entry is *nonzero*; a zero entry indicates a missing edge. Prove that $\theta_{ij} = 0$ implies that X_i and X_j are conditionally independent given all the other coordinates.

Sol.

- (1) Let $p_{U,W}$ be the joint density and let $p_W(w) = \int p_{U,W}(u, w) du$. For $0 < p_W(w) < \infty$, define

$$m(w) = \frac{\int \varphi(u)p_{U,W}(u, w) du}{p_W(w)}$$

whenever the numerator is absolutely integrable, and set $m(w) = 0$ on the exceptional set. By Fubini's theorem and $\mathbb{E}|\varphi(U)| < \infty$, this exceptional set has probability zero under the law of W .

The function m is measurable and

$$\int |m(w)|p_W(w) dw \leq \iint |\varphi(u)|p_{U,W}(u, w) du dw < \infty.$$

For every Borel set C in the state space of W ,

$$\begin{aligned}\mathbb{E}[m(W)\mathbf{1}_{\{W \in C\}}] &= \int_C m(w)p_W(w)dw \\ &= \int_C \int \varphi(u)p_{U,W}(u,w)du dw \\ &= \mathbb{E}[\varphi(U)\mathbf{1}_{\{W \in C\}}].\end{aligned}$$

Hence $m(W)$ is a version of $\mathbb{E}[\varphi(U) \mid W]$, and the claimed conditional-density formula follows.

- (2) For Borel sets $C, D \subseteq \mathbb{R}$, the tower property and the assumed conditional independence imply

$$\begin{aligned}\mathbb{P}(X \in C, Y \in D \mid Z) &= \mathbb{E}[\mathbf{1}_{\{X \in C\}} \mathbb{P}(Y \in D \mid X, Z) \mid Z] \\ &= \mathbb{P}(X \in C \mid Z) \mathbb{P}(Y \in D \mid Z).\end{aligned}$$

By part (1), the joint conditional distribution and the two marginal conditional distributions have densities obtained by dividing their respective joint densities with Z by $f_Z(z)$, for almost every z under the law of Z . First take $C = (-\infty, a]$ and $D = (-\infty, b]$, where $a, b \in \mathbb{Q}$. There are only countably many such rectangles, so outside one null set of z the displayed identity holds for all of them simultaneously. These rectangles determine probability measures on $\mathbb{R}^2$. Thus, for each such z , the conditional joint law equals the product of its conditional marginal laws. By uniqueness of Lebesgue densities,

$$f_{X,Y|Z}(x, y \mid z) = f_{X|Z}(x \mid z) f_{Y|Z}(y \mid z)$$

for Lebesgue-almost every (x, y) , as required.

- (3) For $A \in \sigma(X)$ and $B \in \sigma(Y)$, $\mathbf{1}_A$ is $\sigma(X, Z)$ -measurable. Therefore

$$\begin{aligned}\mathbb{P}(A \cap B \mid Z) &= \mathbb{E}[\mathbb{E}[\mathbf{1}_A \mathbf{1}_B \mid X, Z] \mid Z] \\ &= \mathbb{E}[\mathbf{1}_A \mathbb{P}(B \mid X, Z) \mid Z] \\ &= \mathbb{E}[\mathbf{1}_A \mathbb{P}(B \mid Z) \mid Z] \\ &= \mathbb{P}(A \mid Z) \mathbb{P}(B \mid Z) \quad \text{a.s.}\end{aligned}$$

No density assumption is used.

- (4) By permuting coordinates, the recalled Gaussian conditioning formula applies to any coordinate X_i conditional on the remaining vector X_{-i} . The block inverse formula gives the conditional variance $1/\theta_{ii}$, while the recalled identity for the off-diagonal precision block gives the conditional mean. Consequently,

$$X_i \mid X_{-i} = x_{-i} \sim N\left(-\frac{1}{\theta_{ii}} \sum_{k \neq i} \theta_{ik} x_k, \frac{1}{\theta_{ii}}\right).$$

Here $\theta_{ii} > 0$ because Θ is positive definite. Suppose $\theta_{ij} = 0$, and write $R = (X_k)_{k \neq i, j}$. The displayed conditional law then depends only on R , not on X_j . For each Borel set $D \subseteq \mathbb{R}$, denote its probability under this normal law by $K_D(R)$. Thus

$$\mathbb{P}(X_i \in D \mid X_j, R) = K_D(R).$$

Taking conditional expectations given R yields $\mathbb{P}(X_i \in D \mid R) = K_D(R)$. For every Borel set

$C \subseteq \mathbb{R}$, it follows that

$$\begin{aligned}\mathbb{P}(X_j \in C, X_i \in D \mid R) &= \mathbb{E}[\mathbf{1}_{\{X_j \in C\}} K_D(R) \mid R] \\ &= \mathbb{P}(X_j \in C \mid R) \mathbb{P}(X_i \in D \mid R).\end{aligned}$$

This is conditional independence of X_i and X_j given all the other coordinates. When $p = 2$, R is empty and the same argument gives ordinary independence.